Certificates for a finite unit-distance obstruction in a cyclotomic module

Abstract

This supplement specifies the exact finite evidence underlying the companion construction, its physical gluing interfaces, and a portable replay procedure. The package contains the seed coordinates and edges, quotient data, compressed shortest words, equality-forest records, and complete local overlap geometry. It verifies the retained bounds without expanding the final graph. The seed terminal-forcing result is an explicit cited dependency; the replay does not provide a new satisfiability refutation of that result.

1 Exact representation and the imported premise

All coordinates use one declared complex embedding. Set

$$w = e^{\pi i/21}, \quad K = \mathbb{Q}(w), \quad \mathcal{O} = \mathbb{Z}[w], \quad \rho = w^7, \quad \delta = 2\rho - 1, \quad t = N(w)/7,$$

where

$$\Phi_{42}(X) = X^{12} + X^{11} - X^9 - X^8 + X^6 - X^4 - X^3 + X + 1,$$
$$N(X) = 2 + X + 3X^2 - 2X^4 + X^5 + 5X^6 + X^7 + X^8 - 6X^9 - 4X^{10} + 4X^{11}.$$

Write $M_1 = \mathcal{O} + t\mathcal{O}$ and $M_2 = \mathcal{O} + t\mathcal{O} + t^2\mathcal{O}$. A stored twelve-tuple c represents $\sum_{i=0}^{11} c_i w^i/49$, except that the original seed tuples have denominator 7. Equality is coefficient equality in this fixed power basis. Multiplication is reduction modulo Φ_{42}, and conjugation substitutes w^{-1}. Thus two numerator-49 points at difference d are a unit apart exactly when

$$d\overline{d} = (2401, 0, \ldots, 0)$$

in the integral coefficient representation. There is no floating tolerance.

The indispensable external premise is Haugland's computational terminal-forcing result for the preprint's G_1: every proper four-colouring assigns different colours to 0 and δ [3, Section 3]. The retained seed has 740 distinct points in M_1 and 3985 complete unit edges. Its exact coordinates, edges and distinguished vertices are in `data/BASE_G1.json`. The cited source and the geometry correspondence are different obligations. The source's mathematical forcing assertion is used by citation. Seed-to-source correspondence, membership in M_1, and completeness of the source edge list use the earlier path reconstruction and exact-distance calculation. The companion checks the supplied seed images and their selected edges; it does not repeat that source reconstruction or supply a source SAT refutation.

For $u = t^j w^r$, $0 \le j \le 2$, $0 \le r < 42$, the seed transport is

$$T_u(g) = \begin{cases} ug, & j < 2, \\ -u\overline{g}, & j = 2. \end{cases}$$

The exact identity $N(w)\overline{N(w)} = 49$ makes these maps isometries, and $\overline{\delta} = -\delta$ gives $T_u(\delta) = u\delta$. The third-shell reflection keeps the image in $t^2\mathcal{O} + t\mathcal{O} \subset M_2$. For $A = x$, $B = x + 2u$, $C = x + u$, $D = x + \rho u$, $E = x + (1 + \rho)u$, the gadget $H(x, u)$ uses the seven edges

$$CD, CE, DE, AC, AD, BC, BE$$

and the two complete seed images $A + T_{\rho^{-1}u}(G_1)$ and $D + T_{-\rho u}(G_1)$. The imported inequalities are $A \neq E$ and $D \neq B$. The triangle CDE then leaves the same fourth colour for A and B. This is the two-copy equality mechanism of the cited construction, with the specified transports for this module. It establishes only

$$R_4(H, \{A, B\}) \subseteq \mathrm{EQ} = \{(c, c) : c \in \{0, 1, 2, 3\}\}.$$

The physical relation may be empty. Equality with EQ for the virtual $K_5 - AB$ constraint pattern does not prove extension through the seed interiors.

2 The quotient and the complete line-cover certificate

The odd denominator identifies $M_2/2M_2$ with $V = \mathcal{O}/2\mathcal{O} = \mathbb{F}_2[X]/(\Phi_{42})$. This is a 12-dimensional vector space; it is not being called a field. The integer identifier $a = \sum_i a_i 2^i$ denotes $\sum_i a_i X^i$, and the section is $s(a) = \sum_i a_i w^i$. We write $[z]$ for a residue class modulo $2M_2$; overbars continue to mean complex conjugation. The 63 distinct nonzero residues of $t^j w^r$, $0 \leq r < 21$, split into 21 triples under multiplication by $[\rho]$. The frozen ordering is the lexicographic ordering of the triples shown below. All indices are zero-based.

i	L_i	i	L_i	i	L_i
0	$1, 128, 129$	7	$207, 2595, 2796$	14	$885, 3141, 3888$
1	$2, 256, 258$	8	$243, 3908, 4023$	15	$972, 2154, 2982$
2	$4, 512, 516$	9	$414, 3715, 3869$	16	$977, 1339, 1770$
3	$8, 1024, 1032$	10	$486, 1077, 1491$	17	$1398, 2236, 3530$
4	$16, 2048, 2064$	11	$559, 3312, 3807$	18	$1656, 2754, 3258$
5	$32, 2907, 2939$	12	$699, 1118, 1765$	19	$1944, 2959, 3095$
6	$64, 3501, 3565$	13	$828, 1377, 1629$	20	$1954, 2678, 3540$

For a nonzero binary character mask a, let H_a be the set of lines all of whose vectors have even dot product with a. A deleted-line set D leaves a hitting family precisely when $H_a \not\subseteq D$ for every nonzero a. The checker constructs all 4095 masks H_a, reduces the 1288 distinct masks by inclusion, and obtains 63 minimal pairs and 133 minimal triples. It enumerates every set avoiding those forbidden subsets by adding indices in increasing order. There are 379 such sets, with cardinality counts

$\lvert D\rvert$	0	1	2	3
number	1	21	147	210.

Every allowed increasing extension is present; there are 378 such extensions. This establishes completeness: any missing nonempty allowed set would have an allowed predecessor obtained by deleting its largest element. As a further bounded check, each of the $\binom{21}{4} = 5985$ four-line deletions contains a forbidden mask. The full 379-mask list is stored in `expected/LINE_COVER.json`, not replaced by its cardinality or digest.

For m retained whole lines, the character eigenvalues are $4b_a - m$, where b_a counts retained lines in H_a, and the degree is $3m$. The main paper proves the hitting/non-four-colourability equivalence using this calculation, the ratio bound [2, Theorem 1], and the converse additive colouring $v \mapsto (a(v), a([\rho]v))$ when a character misses all retained lines. Consequently 18 is the exact whole-line threshold in this fixed family. This is not minimality among arbitrary subgraphs or direction families.

The final construction deletes L_5, L_{15}, L_{17}. Its exact 54 residues are bound in `data/COMPARISON.json`; shell, exponent and integral numerator records are in `data/CERTIFICATE.json`. Its spectrum is

λ	-14	-10	-6	-2	2	6	10	14	18	54
multiplicity	135	477	693	864	693	558	459	180	36	1.

Thus it has 4096 vertices, 110592 edges, and independence bound $14336/17 < 1024$. The illustrative line-cover witness instead deletes L_1, L_7, L_8. Its degree and least eigenvalue agree, but its full spectrum differs. Both witnesses are retained, and the replay checks their respective roles rather than substituting one for the other.

3 Root words, order choices and exact equality forests

The root-word problem is the established cyclotomic word metric [1]. The main paper derives a particular seven-pair specialization without asserting its priority. With $\zeta = w^6$, write $n = \sum_{k=0}^{6}(a_k + b_k\rho)\zeta^k$. All representations differ by a common integer shift (A, B), and the shortest length is

$$\min_{A,B\in\mathbb{Z}} \sum_{k=0}^{6} \max\{|a_k + A|, |b_k + B|, |a_k + b_k + A + B|\}.$$

The original generator evaluated at most 147 arrangement vertices and chose the lexicographically first minimizing shift. Each record in `data/ROOT_WORDS.json` stores a twelve-coefficient target, a shift, and 42 nonnegative letter multiplicities. There are 2835 distinct targets. These stored multiplicities define the certificate expansion. The original generator fixed the lexicographic shift and canonical local decomposition; the replay does not separately certify that tie-break, the decomposition rule, or the recorded candidate count. It checks every word sum exactly and uses a different optimality oracle: a bounded integer scan in A, followed by binary search in B for the minimum of a convex integer sequence. A candidate of cost at most L must satisfy $|a_k + A|, |b_k + B| \le L$ for every k, supplying finite scan bounds. The retained independent check used 339332 objective evaluations and maximum A-interval width 115. Its finite minimum check complements, rather than replaces, the main paper's uniform integral-kernel proof.

For an oriented quotient edge $a < b = a \operatorname{xor} [u]$, put $u = t^j w^r$, $d = 7^j$, $U = du$, $k = (d-1)/2$, and $n = (s(a) - s(b) + U)/2 \in \mathcal{O}$. The checked endpoint identity is

$$(s(a) + u - s(b))/2 = n - ku.$$

A root word for n, followed by k letters $-u$, gives a path of doubled unit steps from $s(b)$ to $s(a) + u$. The forest instance uses the retained direction-record order for the chosen residues. Within each fibre b, it compares exactly four root-letter orders:

1. the exponents $0, 1, \ldots, 41$;
2. that list in reverse;
3. the list $(6k + 7h + 21\varepsilon) \bmod 42$, with nested loops $k = 0, \ldots, 6$, $h = 0, 1, 2$, $\varepsilon = 0, 1$;
4. the third list in reverse.

The multiplicities and the $-u$ tail are fixed before this comparison. Choose the order giving the fewest prefix-trie edges, breaking ties by the lowest order index. The independent replay builds actual dictionary tries; the author generator used sorted-word longest common prefixes.

Start the fibre with the point $s(b)$ in `seen`. Walk the prescribed paths. At a step $x \to y = x + 2v$, retain the canonical $H(x, v)$ exactly when y is new, then insert y. If y was already seen, continue the path without adding a gadget. Inductively, the current point is connected to $s(b)$, and a new point attaches by one edge. The retained skeleton is a tree connecting every required endpoint. The independent check reconstructs it and also performs a breadth-first traversal. Different fibres have different residues; this separates their skeleton points, not the interiors of their gadgets.

Quantity in the fixed final recipe	Exact value
Lifted quotient edges and checked endpoints	110592
Independent shortened full-word occurrences	3649656
Shared prefix-trie occurrences	3439213
Retained exact-coordinate forest edges T	3437445
Maximum disjoint eligible same-direction pairs P	1460199
Maximum shortest root word / full word with tail	66 / 90
Maximum anchor or retained endpoint numerator	3087

The file `data/SHORT_FOREST.json` contains all 4096 fibre records, the four orders, per-order prefix totals and the final counts. A retained stream record is the JSON array $[b, [c_0, \ldots, c_{11}], j, r]$, with compact separators and one terminating newline, in the deterministic traversal order. The SHA256 digest binds that stream; the replay regenerates its contents and checks connectivity and counts independently of the digest. The earlier stream-consecutive scan found 1450915 disjoint pairs. For the sharper bound, make an auxiliary graph on retained occurrences in each fibre, joining $H(x, u)$ to $H(x + 2u, u)$ exactly when their indexed directed (j, r) agrees and their endpoints and anchors are equal as twelve-integer coordinate tuples. An occurrence has at most one predecessor and one successor. A cycle would imply $2nu = 0$ for some $n > 0$, so every component is a path; taking $\lfloor m/2 \rfloor$ pairs from each path of length m is a maximum disjoint matching in this eligibility graph. The portable matching record stores per-fibre totals, the path-length histogram, a digest of the matching stream, and a nonconsecutive witness in fibre 256 at stream indices 0 and 12. The separate component traversal reconstructs the eligibility components and verifies this count. These pairs are a packaging of the same frozen forest, not new graph edges or gadgets.

For a global union bound, first count the equality skeleton: it has $4096 + T$ vertices. Each singleton adds at most $1066 - 2 = 1064$ vertices outside the skeleton, and each paired union at most $2045 - 3 = 2042$. Selected edges obey the local union counts. Thus the unpaired and paired vertex bounds are respectively

$$4096 + 1065T = 3660883021,$$
$$4096 + 1065T - 86P = 3535305907,$$

and the selected-edge bound is

$$110592 + 6071T - 198P = 20579719785.$$

These are union bounds, allowing further physical coincidences. Shortest-root optimality concerns the integral n only; it does not optimize the tail, section, quotient, four-order choice or final network.

4 Physical union geometry and complete interfaces

For each $j = 0, 1, 2$, the local certificate explicitly materializes $H(0, t^j) \cup H(2t^j, t^j)$. Each occurrence has 1066 vertices and 6071 selected edges. Haugland's 6264 edges for the source G_2 are induced unit edges [3, Section 3]; our count uses the selected edges specified above. The following counts agree in all three shells.

Union vertices	2045	Shared vertices	87
Selected union edges	11944	Shared selected edges	198
Induced unit edges	12697	Extra unit edges	753
Strict cross edges	367	Vertex saving per pair	86

Here a strict cross edge joins the exclusive vertex sets of the two occurrences. The other extra edges may have both endpoints in one occurrence. The file `data/CHAIN_GEOMETRY.json` stores every physical coordinate, local vertex map, selected and induced edge, shared point, boundary and implication DAG. The replay reconstructs the seed images, checks every selected edge, and tests all pairs of the 2045 union coordinates by exact norm arithmetic. It is this complete local geometry, together with $H(x, t^j w^r) = x + w^r H(0, t^j)$, that justifies uniform directed-pair savings. Reversing an equality statement alone would not justify the same placement.

The distinction between necessary equality implications and exact boundary relations can be expressed precisely. Identify occurrence vertices with the same coordinate before forming any relation. For pieces (V_i, E_i) with union vertex set V, let E_{extra} contain the intended edges absent from the piece edge lists, and let $Z \subseteq V$ be the output vertices. Choose B_i to include $V_i \cap Z$, all intersections $V_i \cap V_h$ for $h \neq i$, and every vertex in V_i incident to an edge of E_{extra}. Let R_i be the exact set of colour assignments on B_i extending to a proper four-colouring of the whole piece. With all variables indexed by physical vertices,

$$R_4(V, \textstyle\bigcup_i E_i \cup E_{\text{extra}}; Z) = \pi_Z \left((\bowtie_i R_i) \bowtie (\bowtie_{pq \in E_{\text{extra}}} \{c : c_p \neq c_q\}) \right). \tag{1}$$

In this notation each join is the compatibility join on common vertex variables; the juxtaposition of the two joins imposes both collections of factors.

Proof. A global colouring restricts to a tuple in every factor. Conversely, a tuple in their join has a witness colouring for each complete boundary assignment. Any vertex shared by two pieces belongs to both boundaries, so these witnesses agree on every overlap and unite into a function on V. Piece edges are proper by their witnesses and remaining edges by the inequality factors. Projection existentially chooses the discarded colours. Larger boundaries preserve the argument. Forgetting a variable before its last overlap or unprocessed incident edge does not. □

For the selected two-piece union E_{extra} is empty; for its induced unit graph it consists of all 753 missing unit edges. The stored induced boundaries include endpoints of missing internal edges as well as cross edges. These data certify incidence and the hypotheses of (1); they do not tabulate R_i. An exact symbolic encoding uses four Boolean variables per physical vertex, one positive four-literal clause and six at-most-one clauses per vertex, and four binary clauses per edge. The two-piece encoding has 8180 variables and 62091 clauses for selected edges, or 65103 clauses for induced edges. These are encoding sizes, not solver results.

For the construction only the weaker implication is needed: any one global colouring is constant on each connected component of the equality skeleton. This follows by restricting it to each actual H and using transitivity. Interior coincidences and extra unit edges can remove colourings but cannot invalidate that implication. Preserving the required pairs in a forest therefore suffices without constructing any local extension witnesses.

5 Implication certificates and finite regression examples

Each stored two-piece DAG has four cited-G_1 leaves, two K_5-minus-edge implications and three equality-rule nodes. It derives end-to-end equality, reverses the second equality, and reuses that same fact in another transitivity step. The replay checks topologically earlier parents, exact endpoints, the 740 source images and 3985 source edges at each leaf, and the seven genuine skeleton edges for each local implication. Symmetry and transitivity must have the declared endpoints. Deliberately changing the final endpoint is rejected in every shell. Reuse of a fact in this DAG does not count its physical gadget twice. The cited forcing implication remains an imported leaf, not a theorem proved by the Python rule checker.

Palette compression retains the equality partition of the *whole* boundary into at most four blocks. Two labelled tuples with the same partition differ by a global permutation of colours, which also acts on each extension. At a join, enumerate global partitions, restrict to each piece, and canonicalize those restrictions. Independent local numerical colour names cannot simply be matched. On the path $a - x - b$, each edge has canonical tuple $(0, 1)$, and literal matching would incorrectly compare $x = 1$ with $x = 0$. The correct global partitions are $(0, 1, 0)$ and $(0, 1, 2)$, yielding 36 labelled colourings and all 16 terminal tuples.

The tiny-relation certificate includes an induced star with centre 0 and leaves $1, -1, (5 + \sqrt{-11})/6, -(5 + \sqrt{-11})/6$. This example belongs to $\mathbb{Q}(\sqrt{-11})$; membership in M_2 is not asserted. Exact squared distances verify the star. Its four-leaf relation has 232 tuples and 14 equality partitions, while all projections onto at most three leaves are full. Joining a three-leaf piece and a one-leaf piece after prematurely forgetting the centre introduces precisely the 24 rainbow tuples. The certificate stores one complete-colouring witness for each accepted full leaf tuple.

Two smaller examples lie in M_2: two edgeless ports both at coordinate 0 have only four physical assignments, instead of sixteen assignments to unidentified names; ports at 0 and 1 have twelve assignments after imposing their unit edge, instead of sixteen when it is omitted. Finally, exhaustive enumeration of the virtual $K_5 - AB$ gives 24 colourings and terminal relation EQ, and a virtual two-piece chain gives 144 colourings. These fixtures check distinct interface errors. None computes the relation or establishes nonemptiness of the actual physical H.

6 Portable replay and coordinate-height verification

The Parallel Attack computational companion, version `companions-v1`, supplies the finite data and checkers. With Python 3.9 or later, from its `computational/` directory, run

```
python3 -E -s -B verify.py -output /path/to/empty-results
```

The standard library is sufficient. The output directory must be empty, and optimized Python execution is rejected because assertions are essential. The verifier resolves inputs relative to its own location and checks their manifest. It runs five programs for the finite construction, maximum eligible matching, a separate matching-component check, complete line-cover enumeration, and tiny relation fixtures. It also verifies the coordinate-height bounds.

The construction checker recomputes words, forests, the earlier consecutive-pair count, spectrum and local geometry. The matching checkers recompute the maximum same-direction path matching and validate its components on the same forest. The construction checker's arithmetic is separately written from the original generators: monomial remainders construct multiplication, and a Laurent autocorrelation formula computes norms. Repeating the supplied implementation is a fresh execution of that implementation, not another implementation or another proof of the cited seed premise.

A separately documented bounded function then checks all 126 generating units and all $126 \cdot 740$ transported seed offsets. In numerator-49 coordinates their maximum absolute coefficients are 63 and 154. Every point of an H occurrence lies within coefficient distance

$$\max(2 \cdot 63,\ 63 + 154) = 217$$

of its anchor. The forest replay checks every retained edge's anchor and new endpoint, attaining maximum 3087. Every visited point was either a section representative or first inserted as such an endpoint, so omitted steps introduce no unchecked anchor. Hence all final coordinates satisfy $|c_i| \leq 3304$ with denominator 49. The bounded offset function uses the construction checker's exact ring functions. Module membership follows from the symbolic transports and paths, not from a coefficient-height test alone.

The computed scientific fields are compared with the supplied expected results. The output directory receives the check results and a concise `RESULT.json`; input hashes are checked again after execution. A hash identifies the input bytes, while the finite mathematical checks establish the assertions described above.

7 Companion files and interpretation

Paths in the following table are relative to the companion's `computational/` directory. The manifest records each supplied program, data file and expected result. The five checker programs are in `checks/`; `verify.py` runs them together.

Artifact	Scientific role
`data/BASE_G1.json`	Exact seed coordinates, complete edge list, terminals.
`data/spectrum.json`; `expected/LINE_COVER.json`	Ordered lines, binary field-action data, all forbidden and allowed deletion masks.
`data/CERTIFICATE.json`; `data/COMPARISON.json`	Direction transports and the chosen final quotient.
`data/ROOT_WORDS.json`	All 2835 integral targets and shortest multiplicities.
`data/SHORT_FOREST.json`	All fibre choices, counts, fixed orders and stream identity.
`expected/MAX_PAIRS.json`	Per-fibre maximum eligible matching counts, path histogram, matching digest and nonconsecutive witness.
`data/CHAIN_GEOMETRY.json`	Three complete physical unions, edges, boundaries and implication DAGs.
`expected/RELATIONS.json`	Exact tiny relations, partitions and extension witnesses.
`expected/FINITE_CHECK_RESULTS.json`	Expected finite construction results for comparison.

The certificate requires the seed, quotient data, words, forest, local geometry and checking programs together. The finite recipe can be checked without producing an explicit billion-vertex coordinate list. Full expansion would have output and identification costs proportional to its occurrences and edges; compressed root-word counts likewise do not remove the cost of writing expanded words.

The source forcing result remains cited; the whole-line threshold concerns precisely the fixed 21-line family; the root-word minimum concerns the fixed integral subproblem. The selected graph has no proper

four-colouring under these dependencies, and its induced unit supergraph has the same obstruction. The displayed edge bound applies only to the selected graph. No exact five-colouring, minimum obstruction, global optimization or literature size record is certified here. The established word-metric framework is credited to Beck–Hosten; priority of the explicit specialization and this module construction remains unresolved by the retained bounded comparison.

References

[1] Matthias Beck and Serkan Hoşten. Cyclotomic polytopes and growth series of cyclotomic lattices, 2006. arXiv:math/0508136v4, Theorems 3 and 13.

[2] Willem H. Haemers. Hoffman's ratio bound, 2021. arXiv:2102.05529v2, Theorem 1.

[3] Jan Kristian Haugland. A Moser-spindle-free 5-chromatic unit distance graph on 2131 vertices in the plane, 2026. arXiv:2608.04542v5, Section 3.